

数学物理方程(理科版)

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第四章 位势方程

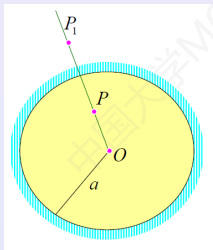
§1. 基本解与Green函数

§1.3 圆上的Poisson公式

§1.3 圆上的Poisson公式

设 B_a 为以原点为心, a 为半径的开圆盘. 下面用Green函数法, 求解圆盘上位势方程的Dirichlet边值问题:

$$\begin{cases} -\Delta u = f(x), & (x, y) \in B_a; \\ u = \varphi, & (x, y) \in \partial B_a, \end{cases}$$



解: 第一步, 求Green函数

对任意的 $P(\xi, \eta) \in B_a$, 记 $P_1(\xi_1, \eta_1)$ 是其关于圆周 ∂B_a 的对称点, 则 P_1 满足:

- ① 位于射线 OP 上
- ② $\overline{OP} \cdot \overline{OP_1} = a^2$.

若 P 正好是原点, 则它的对称点为无穷远点.

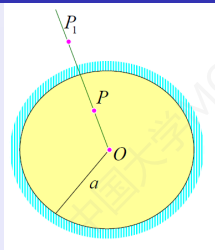
镜像法求Green函数

(1) 如果 $P(\xi, \eta)$ 正好是坐标原点, 则当 $(x, y) \in \partial B_a$ 时,

$$\Gamma(x, y; \xi, \eta) = -\frac{1}{2\pi} \ln \sqrt{(x - \xi)^2 + (y - \eta)^2} = -\frac{1}{2\pi} \ln a,$$

从而 $G(x, y; 0, 0) = \Gamma(x, y; 0, 0) + \frac{1}{2\pi} \ln a = \frac{1}{4\pi} \ln \frac{a^2}{x^2 + y^2}$.

(2) 如果 $P(\xi, \eta) \in B_a \setminus \{(0, 0)\}$, 在 P 放置一个单位正电荷, 产生的电位势分布为 $\Gamma(x, y; \xi, \eta)$. 现要在 $P_1(\xi_1, \eta_1)$ 放置一个负电荷, 使其产生的电位势在 ∂B_a 与 $\Gamma(x, y; \xi, \eta)$ 抵消.



设 $Q(x, y)$ 为 ∂B_a 任意一点(如图),则

$$\Gamma(x, y; \xi, \eta) = -\frac{1}{2\pi} \ln \sqrt{(x - \xi)^2 + (y - \eta)^2} = -\frac{1}{2\pi} \ln \overline{QP},$$

$$\Gamma(x, y; \xi_1, \eta_1) = -\frac{1}{2\pi} \ln \sqrt{(x - \xi_1)^2 + (y - \eta_1)^2} = -\frac{1}{2\pi} \ln \overline{QP_1}.$$

因为 $\overline{QP} \neq \overline{QP_1}$, 所以 $\Gamma(x, y; \xi, \eta) - \Gamma(x, y; \xi_1, \eta_1) \neq 0$.

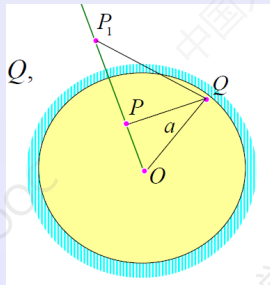
因此我们要对 $\Gamma(x, y; \xi_1, \eta_1)$ 作适当调整.

由于 $\overline{OP} \cdot \overline{OP_1} = a^2$, 所以 $\triangle OQP \sim \triangle OP_1Q$, 从而

$$\frac{\overline{QP}}{\overline{QP_1}} = \frac{\overline{OP}}{\overline{OQ}} = \frac{\overline{OP}}{a},$$

即

$$\overline{QP} = \frac{\overline{OP}}{a} \overline{QP_1}.$$



所以,当 $Q(x, y) \in \partial B_a$ 时, 上式代入得

$$\begin{aligned} 0 &= \left[-\frac{1}{2\pi} \ln \overline{QP} \right] - \left[-\frac{1}{2\pi} \ln \left(\frac{\overline{OP}}{a} \overline{QP_1} \right) \right] \\ &= \left[-\frac{1}{2\pi} \ln \overline{QP} \right] - \left[-\frac{1}{2\pi} \ln \overline{QP_1} \right] + \left[\frac{1}{2\pi} \ln \frac{\overline{OP}}{a} \right] \\ &= \Gamma(x, y; \xi, \eta) - \Gamma(x, y; \xi_1, \eta_1) + \frac{1}{2\pi} \ln \frac{\sqrt{\xi^2 + \eta^2}}{a}. \end{aligned}$$

于是 $\forall Q(x, y) \in B_a$, 有

$$\begin{aligned} G(x, y; \xi, \eta) &= \Gamma(x, y; \xi, \eta) - \Gamma(x, y; \xi_1, \eta_1) + \frac{1}{2\pi} \ln \frac{\sqrt{\xi^2 + \eta^2}}{a}, \\ &= \frac{1}{2\pi} \ln \frac{\sqrt{(x - \xi)^2 + (y - \eta)^2}}{\sqrt{(x - \xi_1)^2 + (y - \eta_1)^2}} + \frac{1}{2\pi} \ln \frac{\sqrt{\xi^2 + \eta^2}}{a}. \end{aligned}$$

$$\Rightarrow G(x, y; \xi, \eta) = \frac{1}{4\pi} \ln \frac{(x - \xi)^2 + (y - \eta)^2}{(x - \xi_1)^2 + (y - \eta_1)^2} + \frac{1}{4\pi} \ln \frac{\xi^2 + \eta^2}{a}.$$

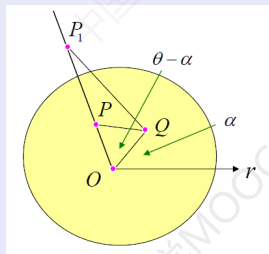
设在极坐标下,

$$Q(x, y) = (r, \alpha), \quad P(\xi, \eta) = (\rho, \theta), \quad P_1(\xi_1, \eta_1) = (\rho_1, \theta), \quad \rho\rho_1 = a^2.$$

由余弦定理,

$$(x - \xi)^2 + (y - \eta)^2 = \overline{QP} = r^2 + \rho^2 - 2r\rho \cos(\theta - \alpha),$$

$$\begin{aligned} (x - \xi_1)^2 + (y - \eta_1)^2 &= \overline{QP_1} = r^2 + \rho_1^2 - 2r\rho_1 \cos(\theta - \alpha), \\ &= r^2 + \frac{a^4}{\rho^2} - 2r\frac{a^2}{\rho} \cos(\theta - \alpha). \end{aligned}$$



圆盘上的Green函数

$$G(r, \alpha; \rho, \theta) = \frac{1}{4\pi} \ln \frac{r^2 + \frac{a^4}{\rho^2} - 2r\frac{a^2}{\rho} \cos(\theta - \alpha)}{r^2 + \rho^2 - 2r\rho \cos(\theta - \alpha)} + \frac{1}{4\pi} \ln \frac{\rho^2}{a^2}$$
$$\Rightarrow G(r, \alpha; \rho, \theta) = \frac{1}{4\pi} \ln \frac{\rho^2 r^2 + a^4 - 2r\rho a^2 \cos(\theta - \alpha)}{a^2[r^2 + \rho^2 - 2r\rho \cos(\theta - \alpha)]}. \quad (*)$$

注意到：上式当 $\rho = 0$,也即 $P(\xi, \eta)$ 正好是坐标原点时,

$$G(r, \alpha; 0, \theta) = \frac{1}{4\pi} \ln \frac{a^2}{r^2} = \frac{1}{4\pi} \ln \frac{a^2}{x^2 + y^2},$$

与前面得到的结果一致. 因此(*)式包含了 $P(\xi, \eta)$ 为坐标原点时的情形.

第二步, 用Green函数求解公式: 由于表示式:

$$u(\xi, \eta) = \iint_{\Omega} G f dx dy - \int_{\partial\Omega} \varphi \frac{\partial G}{\partial \mathbf{n}} ds, \quad \forall (\xi, \eta) \in \Omega,$$

$$\Rightarrow u(\rho, \theta) = \iint_{B_a} G(r, \alpha; \rho, \theta) f(r, \alpha) r dr d\alpha - \int_{\partial B_a} \varphi(r, \alpha) \frac{\partial G(r, \alpha; \rho, \theta)}{\partial \mathbf{n}} ds, \quad \forall (\rho, \theta) \in B_a,$$

其中 $\mathbf{n}$ 为 ∂B_a 上圆 B_a 的**单位外法向量**. 考虑在 ∂B_a 上,

$$\begin{cases} ds = a d\alpha, \\ \frac{\partial}{\partial \mathbf{n}} = \frac{\partial}{\partial r}, \end{cases}$$

所以

$$\begin{aligned}\left. \frac{\partial G(r, \alpha; \rho, \theta)}{\partial r} \right|_{r=a} &= \left. \frac{\partial}{\partial r} \left\{ \frac{1}{4\pi} \ln \frac{\rho^2 r^2 + a^4 - 2r\rho a^2 \cos(\theta - \alpha)}{a^2[r^2 + \rho^2 - 2r\rho \cos(\theta - \alpha)]} \right\} \right|_{r=a} \\&= \frac{1}{4\pi} \left(\frac{2\rho^2 r - 2\rho a^2 \cos(\theta - \alpha)}{\rho^2 r^2 + a^4 - 2r\rho a^2 \cos(\theta - \alpha)} - \frac{2r - 2\rho \cos(\theta - \alpha)}{r^2 + \rho^2 - 2r\rho \cos(\theta - \alpha)} \right) \Big|_{r=a} \\&= \frac{1}{4\pi} \frac{2\rho^2 a - 2\rho a^2 \cos(\theta - \alpha)}{\rho^2 a^2 + a^4 - 2a\rho a^2 \cos(\theta - \alpha)} - \frac{1}{4\pi} \frac{2a - 2\rho \cos(\theta - \alpha)}{a^2 + \rho^2 - 2a\rho \cos(\theta - \alpha)}, \\&= \frac{1}{2\pi a} \frac{\rho^2 - a^2}{a^2 + \rho^2 - 2a\rho \cos(\theta - \alpha)}.\end{aligned}$$

于是对 $\forall (\rho, \theta) \in B_a$, 其表达式

$$\begin{aligned} u(\rho, \theta) = & \frac{1}{4\pi} \int_0^{2\pi} r dr \int_0^a \ln \frac{\rho^2 r^2 + a^4 - 2r\rho a^2 \cos(\theta - \alpha)}{a^2[r^2 + \rho^2 - 2r\rho \cos(\theta - \alpha)]} f(r, \alpha) d\alpha \\ & + \frac{1}{2\pi} \int_0^{2\pi} \frac{\rho^2 - a^2}{a^2 + \rho^2 - 2a\rho \cos(\theta - \alpha)} \varphi(\alpha) d\alpha \end{aligned}$$

为边值问题:

$$\begin{cases} -\Delta u = f(x, y), & (x, y) \in B_a; \\ u = \varphi(x, y), & (x, y) \in \partial B_a \end{cases}$$

的形式解. 特别地, 当 $f \equiv 0$ 时, 得到边值问题

$$\begin{cases} -\Delta u = 0, & (x, y) \in B_a; \\ u = \varphi, & (x, y) \in \partial B_a \end{cases} \quad (1)$$

解的表达式(称为圆上的Poisson公式):

$$u(\rho, \theta) = \frac{1}{2\pi} \int_0^{2\pi} \frac{\rho^2 - a^2}{a^2 + \rho^2 - 2a\rho \cos(\theta - \alpha)} \varphi(\alpha) d\alpha. \quad (2)$$

定理1.6 设 $\varphi \in C(\partial B_a)$, 则由(2)给出的函数是边值问题(1)的解.

证明: 要证: 1. $u \in C^2(B_a)$, 且 $\Delta u = 0$ 于 B_a ; 2. $\lim_{(\rho, \theta) \rightarrow (a, \theta_0)} u(\rho, \theta) = \varphi(\theta_0), \quad \forall \theta_0 \in [0, 2\pi]$,

也即 $u \in C(\overline{B_a})$ 且满足边界条件.

注1: 求Green函数的镜像法同样适合于高维问题($n \geq 3$);

注2: 对于二维问题还可以用复变函数中的保角变换, 来求Green函数, 且非常有效.

同学们，再见！